



Customer Shopping Behavior Analysis

Uncovering insights from **3,900 transactions** across product categories — spending patterns, segments, preferences, and subscriptions.

Dataset at a Glance

3,900

Purchases

18

Columns

37

Missing Ratings

Key Features

- **Demographics:** Age, Gender, Location, Subscription
- **Purchase:** Item, Category, Amount, Season, Size
- **Behavior:** Discounts, Frequency, Rating, Shipping

Data Preparation in Python

1

Load & Explore

Imported with `pandas`; used `df.info()` and `.describe()`

2

Clean & Standardize

Imputed missing ratings by category median; renamed columns to `snake_case`

3

Feature Engineering

Created `age_group` and `purchase_frequency_days` columns

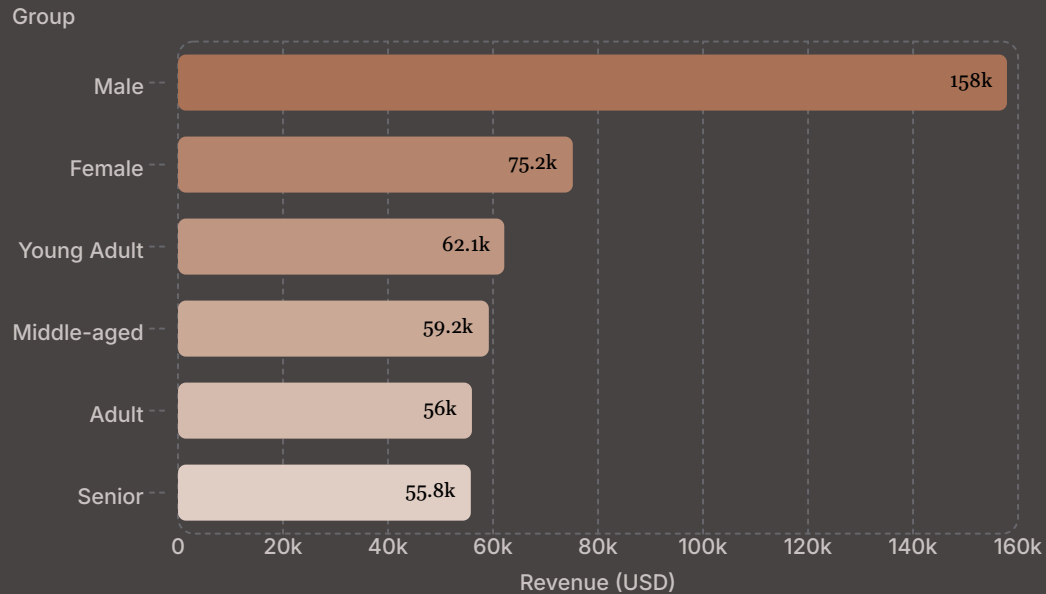
4

Database Load

Connected to PostgreSQL; loaded cleaned DataFrame for SQL analysis



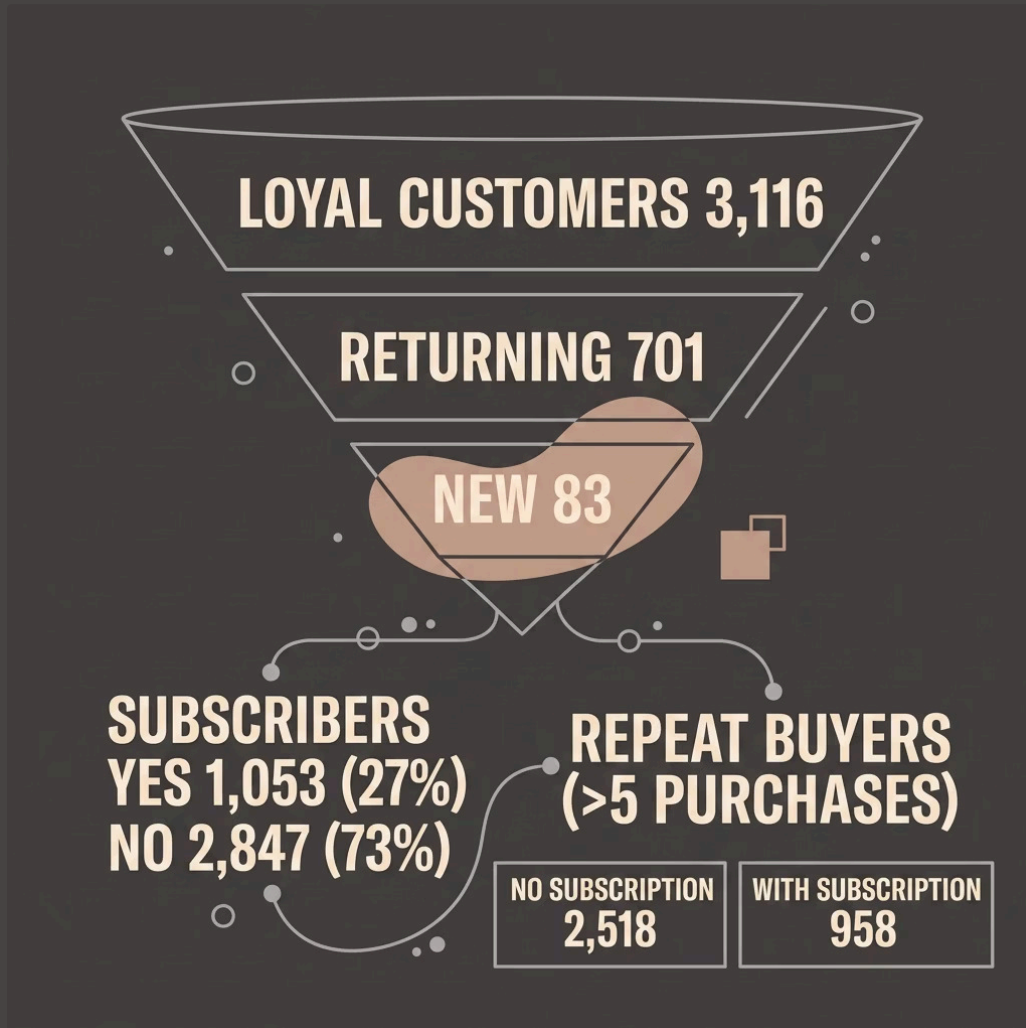
SQL Analysis Highlights



Key Findings

- **Gender:** Male customers generated 2× more revenue than female
- **Age:** Young Adults lead revenue contribution
- **Shipping:** Express avg \$60.48 vs. Standard \$58.46
- **Discount users:** 839 customers spent above average despite discounts
- **Top-rated product:** Gloves (3.86 ★)

Customer Segmentation & Subscriptions



Segment Breakdown

- **Loyal:** 3,116 customers
- **Returning:** 701 customers
- **New:** 83 customers

Subscription Status

- **Subscribers (27%):** avg spend \$59.49, revenue \$62,645
- **Non-subscribers (73%):** avg spend \$59.87, revenue \$170,436

i Most repeat buyers are **not** subscribed — a key growth opportunity.

Power BI Dashboard & Recommendations



Boost Subscriptions

Promote exclusive subscriber benefits

Loyalty Programs

Reward repeat buyers to grow the Loyal segment

Discount Policy

Balance sales boosts with margin control

Targeted Marketing

Focus on high-revenue age groups & express-shipping users